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Experiment 1: Import and Export Data files

Program:

```
import pandas as pd
import os

# ImportDataFiles
base_dir = os.path.dirname(os.path.abspath(__file__))
file_path = os.path.join(base_dir, '..', 'dataset', 'cancer.csv')

df = pd.read_csv(file_path)
print(df.head())

# ExportDataFiles
with open("output/output.txt", "w") as f:
    f.write("Hello World\n")
    f.write(df.head().to_string())
```

Output:

```
oratory/datascience-learning/LabExercises-Sem1/experiments/1ImportExportDataFiles.py
   Stage             Trt  Time                Status  Age   wt                pf  ...                        ekg         hg  sz  sg        ap  bm        Date
0      3  0.2 mg estrogen    72                 alive   75   76    normal activity  ...                heart strain  13.798828   2   8  0.299988   0  1977-08-10
1      3  0.2 mg estrogen     1        dead - other ca   54  116    normal activity  ...  heart block or conduction def  14.599609  42  10  0.699951   0  1977-09-21
2      3  5.0 mg estrogen    40  dead - cerebrovascular   69  102    normal activity  ...                heart strain  13.398438   3   9  0.299988   0  1978-01-12
3      3  0.2 mg estrogen    20  dead - cerebrovascular   75   94  in bed < 50% daytime  ...                      benign  17.597656   4   8  0.899902   0  1978-03-19
4      3         placebo    65                 alive   67   99    normal activity  ...                      normal  13.398438  34   8  0.500000   0  1978-03-22

[5 rows x 17 columns]
```

Experiment2: Measure of Central Tendency

Program

```
import numpy as np
import pandas as pd

ages = [1,2,2,3]

print("Mean: " + str(np.mean(ages)))  # Mean
print("Median: " + str(np.median(ages)))  # Median
print("StandardDeviation: " + str(np.std(ages)))  # Standard Deviation
print("Variance: " + str(np.var(ages)))  # Variance
df  = pd.DataFrame(ages, columns=['ages'])
print("Mode: " + str(df.ages.mode()))  # Mode

q1 = np.percentile(ages, 25)
q3 = np.percentile(ages, 75)

iqr = q3 - q1
print(f"Interquartile Range (IQR): {iqr}")
print("Range: " + str(np.max(ages) - np.min(ages)))  # Range
```

Output:

```
Mean: 2.0
Median: 2.0
StandardDeviation: 0.7071067811865476
Variance: 0.5
Mode: 0    2
Name: ages, dtype: int64
Interquartile Range (IQR): 0.5
Range: 2
```

Without using BuiltIn functions

Program:

```
def find_mean(data):
    n = len(data)
    total = sum(data)
    mean = total / n
    return mean

def find_median(data):
    sorted_data = sorted(data)
    n = len(sorted_data)
    mid = n // 2
```

```
    if n % 2 == 0:
        median = (sorted_data[mid - 1] + sorted_data[mid]) / 2
    else:
        median = sorted_data[mid]
    return median

def find_mode(data):
    frequency = {}
    for value in data:
        frequency[value] = frequency.get(value, 0) + 1

    max_freq = max(frequency.values())
    modes = [key for key, val in frequency.items() if val == max_freq]

    if len(modes) == len(frequency):
        return "No mode"
    elif len(modes) == 1:
        return modes[0]
    else:
        return modes

def find_std_deviation(data):
    mean = find_mean(data)
    squared_diff = [(x - mean) ** 2 for x in data]
    variance = sum(squared_diff) / len(data)
    std_dev = variance ** 0.5
    return std_dev

data = [69, 74, 68, 70, 72, 67, 66, 70, 76, 68, 72, 79, 74, 67, 66, 71, 74]

print("Data:", data)
print("Mean:", find_mean(data))
print("Median:", find_median(data))
print("Mode:", find_mode(data))
print("Standard Deviation:", round(find_std_deviation(data), 2))
```

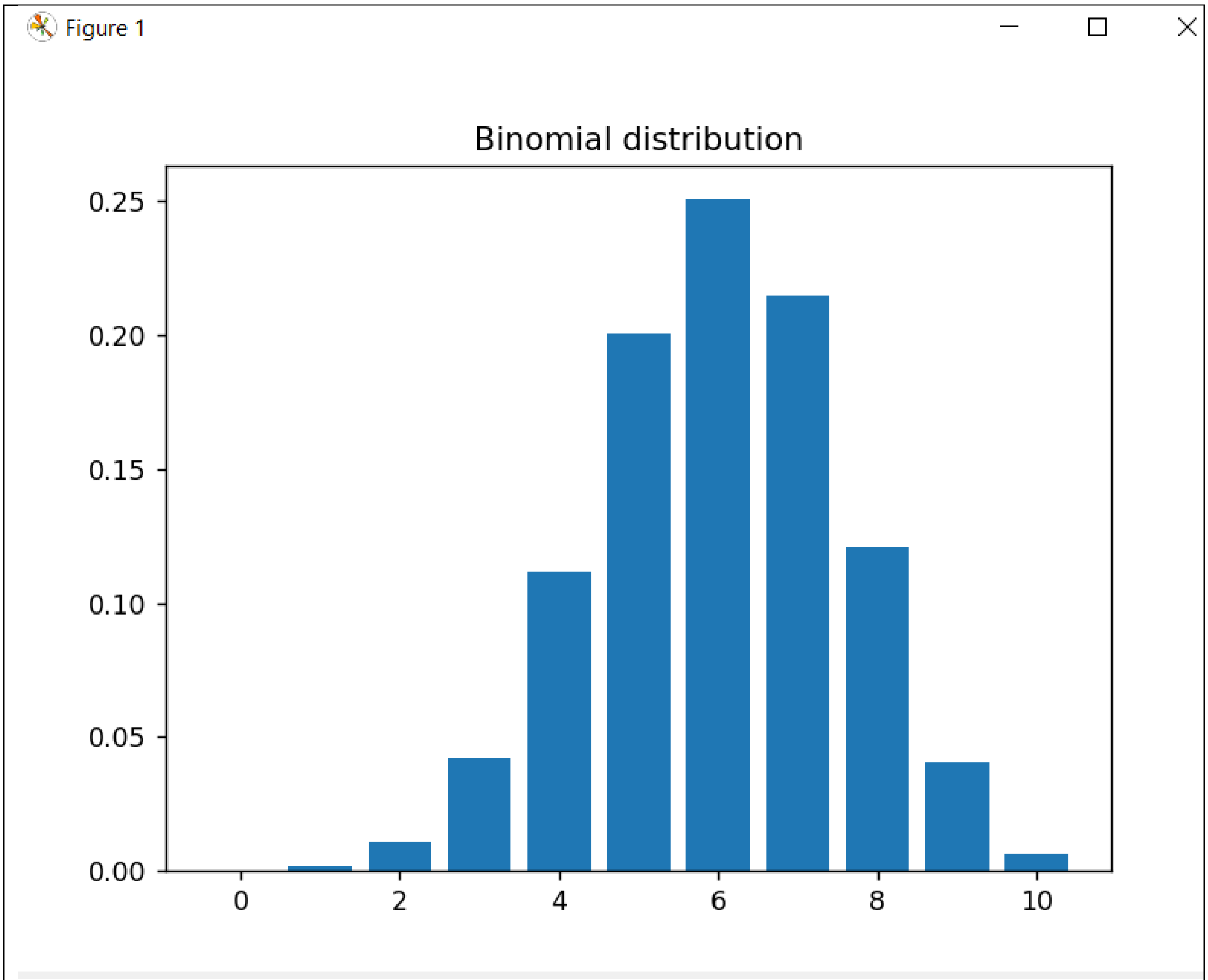
Output

```
Data: [69, 74, 68, 70, 72, 67, 66, 70, 76, 68, 72, 79, 74, 67, 66, 71, 74]
Mean: 70.76470588235294
Median: 70
Mode: 74
Standard Deviation: 3.62
```

Experiment 3: Probability Distributions

```
# Binomial Distribution
from scipy.stats import binom
import matplotlib.pyplot as plt

n = 10
p = 0.6

r_values = list(range(n+1))

dist = [binom.pmf(r,n,p) for r in r_values]

plt.bar(r_values, dist)

_ = plt.title(f'Binomial distribution')

plt.show()
```

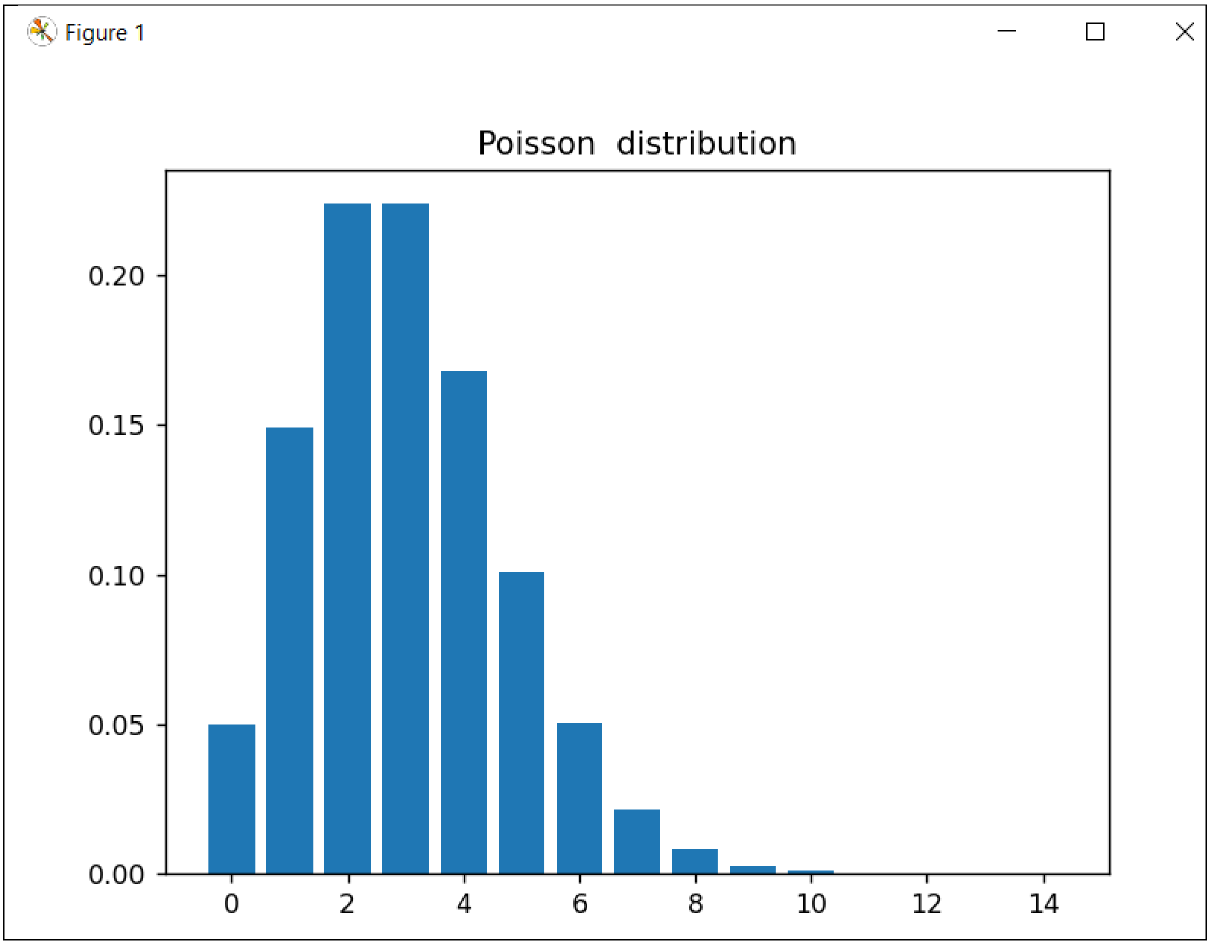
Output



Poisson Distribution

```
from scipy.stats import poisson
import matplotlib.pyplot as plt
import numpy as np

k_values = np.arange(0,15)

pmf_values = poisson.pmf(k_values, mu=3)

plt.bar(k_values,pmf_values)

_ = plt.title(f'Poisson  distribution')

plt.show()
```

Output



Normal Distribution

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

mean = 0
std_dev = 1

x = np.linspace(-4,4,100)

print(x)
y = norm.pdf(x,mean, std_dev)

plt.plot(x,y)
plt.title("Normal Distribution( Mean = 0, SD = 1)")
plt.xlabel("X Values")
plt.ylabel("Probability Density")
plt.show()
```

Output